

Finally, let's look at part d) and answer questions 10 and 11. How does random initialization differ from the trained model we've been inspecting in the previous parts?

Answers to Questions 10 and 11

10. What differences do you notice in the attention patterns between the randomly initialized and trained BERT models?

Random initialization differs significantly from the trained model: In the trained BERT (previous parts), attention patterns are structured, sparse, and meaningful—strong lines connect relevant tokens (e.g., pronouns to antecedents like "it" to "party"), with hierarchical evolution across layers (local in early, global in later). QKV dot products peak for semantic/syntactic pairs, producing focused values.

In the randomly initialized model, patterns are noisy and uniform—lines are faint and evenly distributed across all tokens (no strong connections), reflecting unlearned weights. QKV yields random, flat dot products (~uniform after softmax), lacking focus or hierarchy, as `init_weights()` resets parameters to random values without training.

11. Run the final cell in the notebook. What are some words or tokens that you would expect strong attention between? What might you guess about the gradients of this attention head for those words?

Assuming the final cell visualizes the untrained model on a sentence like "I wanted to have a party [SEP] I bought a cake for it" (consistent with part b), expected strong attention in a trained model would be between "it" and "party" (pronoun resolution), or "cake" and "party" (thematic).

In untrained, it's random—no strong links. Gradients (during training) would be high for these pairs: Backprop would increase weights for relevant QK dot products (positive gradients for "it-party" to learn resolution), decreasing irrelevant ones, shaping uniform to structured patterns via loss minimization.

For question 11, is there a particular token that we would expect to learn high attentions for, based on the results from the previous sections?

Revised Answer to Question 11 (more precise)

Yes — based on everything we saw in parts (a)–(c), the single token we most consistently expect to develop **very strong attention** during training is the pronoun **"it"** attending to its antecedent **"party"**.

- In the trained BERT visualizations (layers 5–11), the line from "it" → "party" is one of the thickest and most persistent connections, even across the [SEP] token.
- In the random-initialization visualization, this connection is just as weak and noisy as everything else.

Therefore, during pre-training (especially on the masked language modeling objective), the gradients for the query of "it" and the key of "party" will be among the largest and most consistent in the model. The attention head will quickly learn to assign a very high attention weight from the pronoun "it" back to "party" (and similar pronoun-antecedent pairs), turning the initially uniform random pattern into the sharp, high-weight edge we observe in the trained model.

So the token pair we expect to see the strongest learned attention (and thus the largest, most directed gradients early in training) is **"it" → "party"**.

What about special tokens?

Updated Answer to Question 11 – Special Tokens

Yes — in addition to the pronoun-antecedent pair ("it" → "party"), the **special token that most consistently develops extremely high attention during training is [CLS]**.

From everything we saw across parts (b) and (c):

- In the **trained BERT** (especially in layers 8–11), the **[CLS] token** receives **very strong incoming attention** from almost every content token** in the sequence. In many heads, you see thick vertical lines converging on [CLS], making it the single most-attended-to token by the end of the network.
- In the **randomly initialized BERT** (part d), attention to [CLS] is completely flat and noisy — just like everything else.

Therefore, during pre-training (particularly the Next-Sentence Prediction objective, but also MLM), the model learns to use [CLS] as a **sentence-level aggregator**. The gradients that flow into the query vectors of normal tokens and the key/value vectors of [CLS] become some of the largest and most

systematic in the entire model. The network quickly turns the initially random attention into a pattern where nearly every token strongly attends to [CLS] in deeper layers.

So the two token relationships we expect to see the **strongest learned attention weights** (and therefore the largest, most consistent gradients early in training) are:

1. “it” → “party” (pronoun coreference)
2. **Almost every token → [CLS]** (sentence representation pooling)

The second one, → [CLS], is actually the most universal and strongest signal the model learns across virtually all inputs during BERT pre-training.

Thanks Grok!

You're welcome! And nice screenshots – looks like you got the notebook running perfectly.

To wrap up everything cleanly for part (d), questions 10 & 11:

10. The randomly initialized BERT shows completely **noisy, uniform, faint attention** everywhere – every token attends roughly equally weakly to every other token (including across [SEP]).

In contrast, the **trained BERT** we examined in parts (a)–(c) has **sharp, structured, hierarchical patterns**: strong lines for syntactic neighbors early, pronoun→antecedent and long-range thematic links in middle layers, and broad convergence on [CLS] in deep layers.

11. In a trained model we consistently see two extremely strong attention signals that are almost completely absent in the random model and therefore receive the **largest, most reliable gradients during pre-training**:

1. “it” → “**party**” (classic pronoun–antecedent coreference) – one of the single strongest individual edges in mid-to-late layers.
2. **Nearly every content token → [CLS]** – in layers 8–11 the [CLS] token becomes the **global attention sink**, receiving thick incoming lines from almost the entire sequence in many heads. This is the most universal high-attention pattern BERT learns, driven heavily by the Next-Sentence Prediction objective.

So the two relationships that flip from “random noise” to “extremely high attention” (and thus see the biggest gradient updates early in training) are:

- **pronouns → their antecedents** (especially “it” → “party” in this example)
- **all tokens → [CLS]** (sentence-level aggregation)

That’s exactly what turns the initially meaningless random weights into the powerful contextual representations we observed in the trained model.

Glad we could walk through the whole thing – great job on the homework! 🚀

↳ Explain NSP objective details

↳ Attention in GPT models